Experiment: Implement a basic Jenkins CI/CD pipeline

Weight: 5 marks | Course outcome: CO1

A Jenkins pipeline job was created that pulls source code from the GitHub repository and triggers an automated build. The first build ran all stages successfully.

Job configuration

- Job name: devops-lab-pipeline (Pipeline type)
- Definition: Pipeline script from SCM -> Git
- Repository URL: https://github.com/theabhinav0231/devops-lab-1.git
- Branch: */main | Script path: Jenkinsfile
- Build trigger: Poll SCM, schedule H/5 * * * *
- Discard old builds: keep last 10

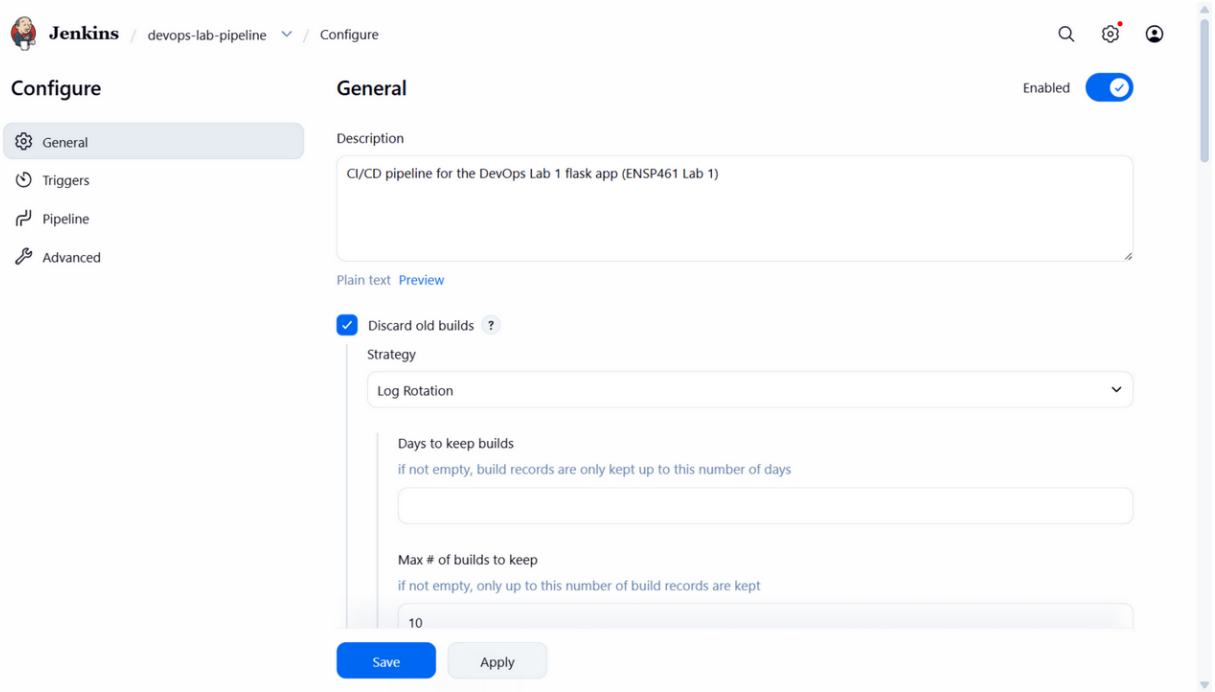


Fig: Pipeline job general configuration

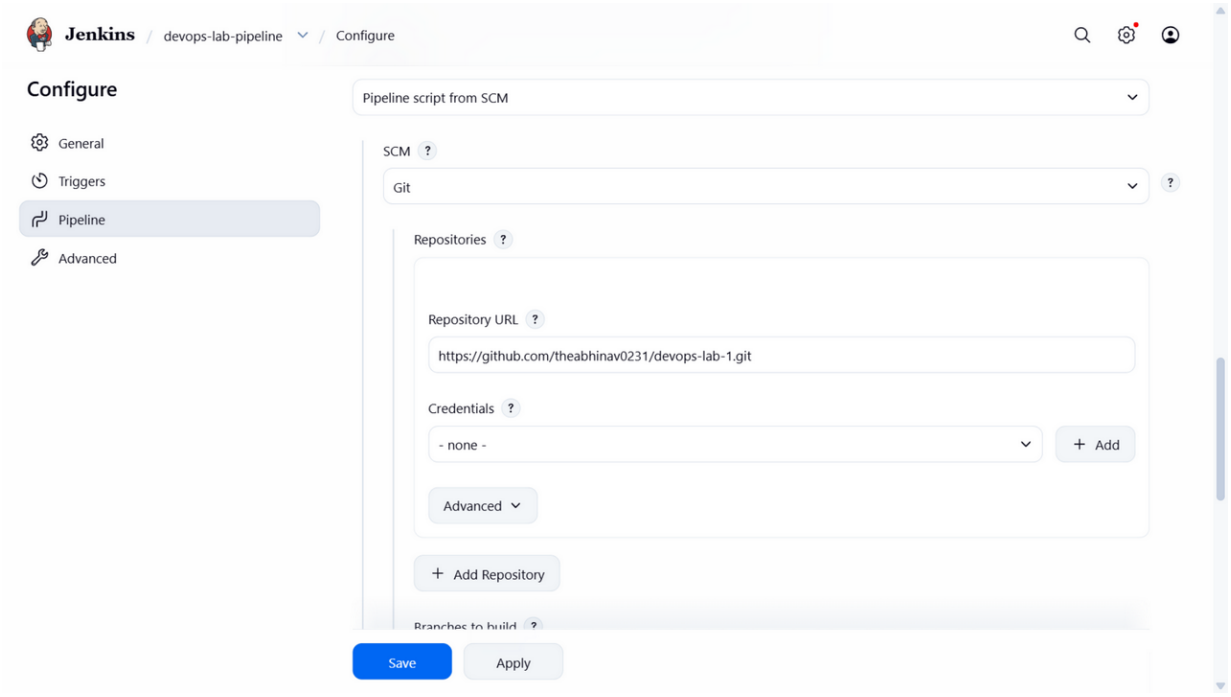


Fig: Pipeline definition from SCM (Git repository URL)

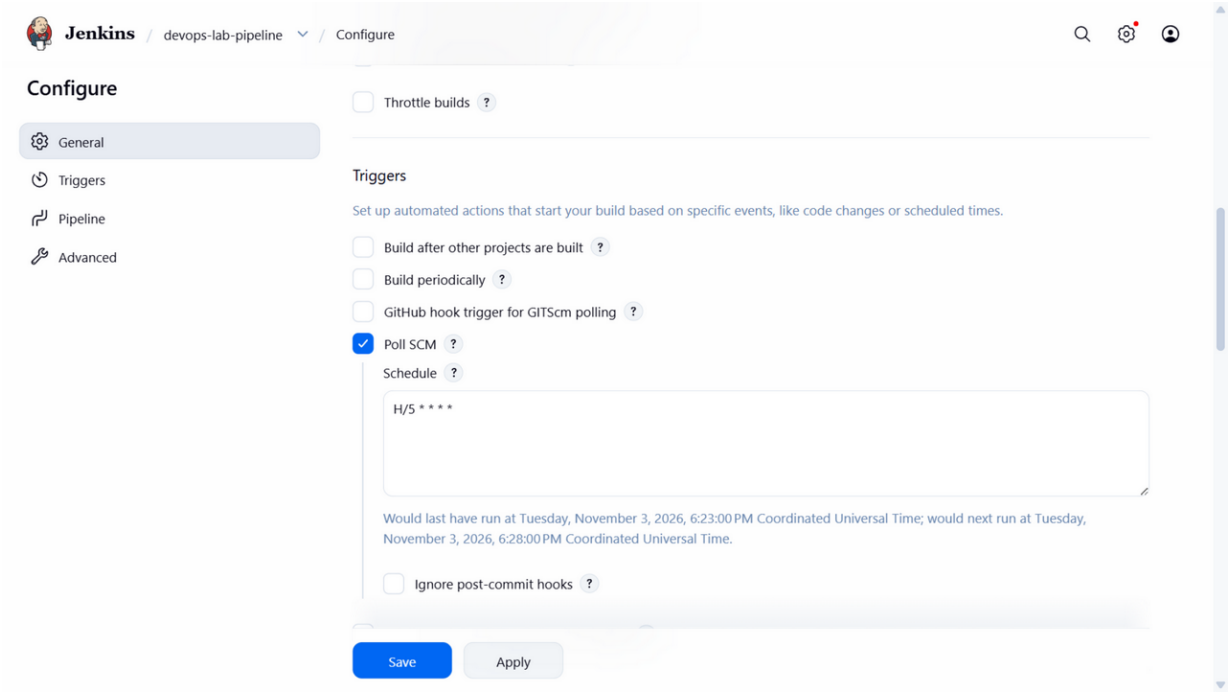


Fig: Poll SCM trigger configured with H/5 * * * *

First automated build

Build Now (build #1) was executed: Jenkins checked out commit 8946b3f from GitHub, installed dependencies, ran pytest, packaged the app and archived the artifact app-build-1.tar.gz. The build finished in 29 seconds.

```
Started by user Abhinav
Obtained Jenkinsfile from git https://github.com/theabhinav0231/devops-lab-1.git
[Pipeline] stage
[Pipeline] { (Checkout)
...
tests/test_app.py::test_index_returns_message PASSED
tests/test_app.py::test_health_returns_up PASSED
[Pipeline] archiveArtifacts
Archiving artifacts
Finished: SUCCESS
```

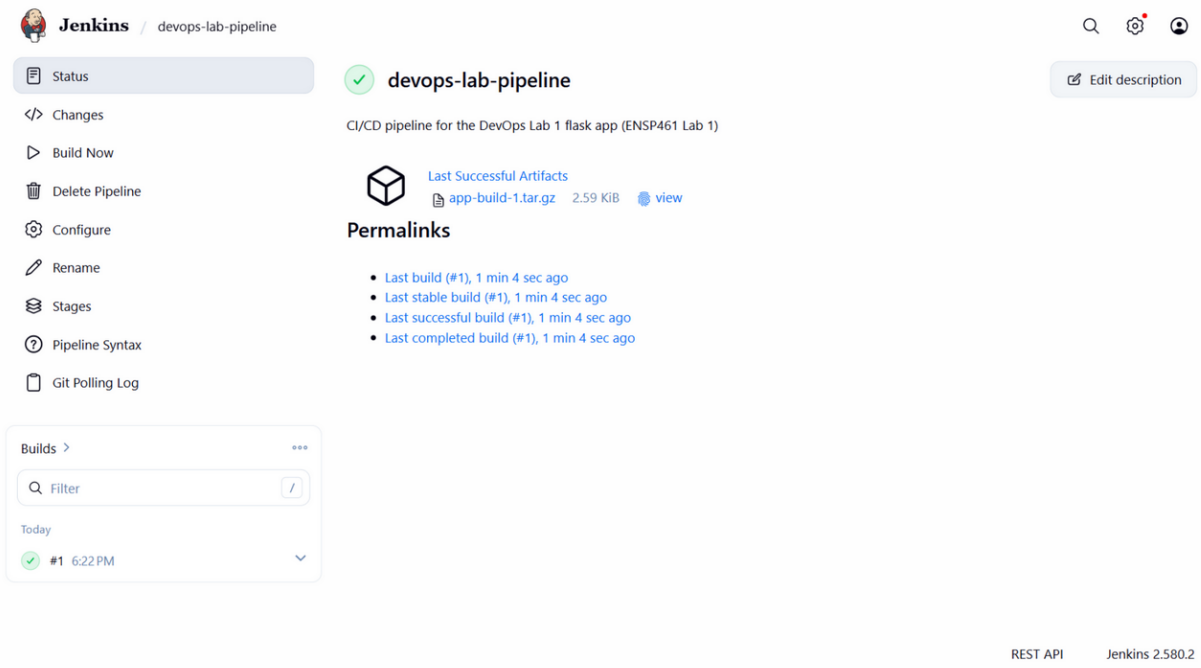


Fig: Pipeline job page - build #1 green, artifact archived

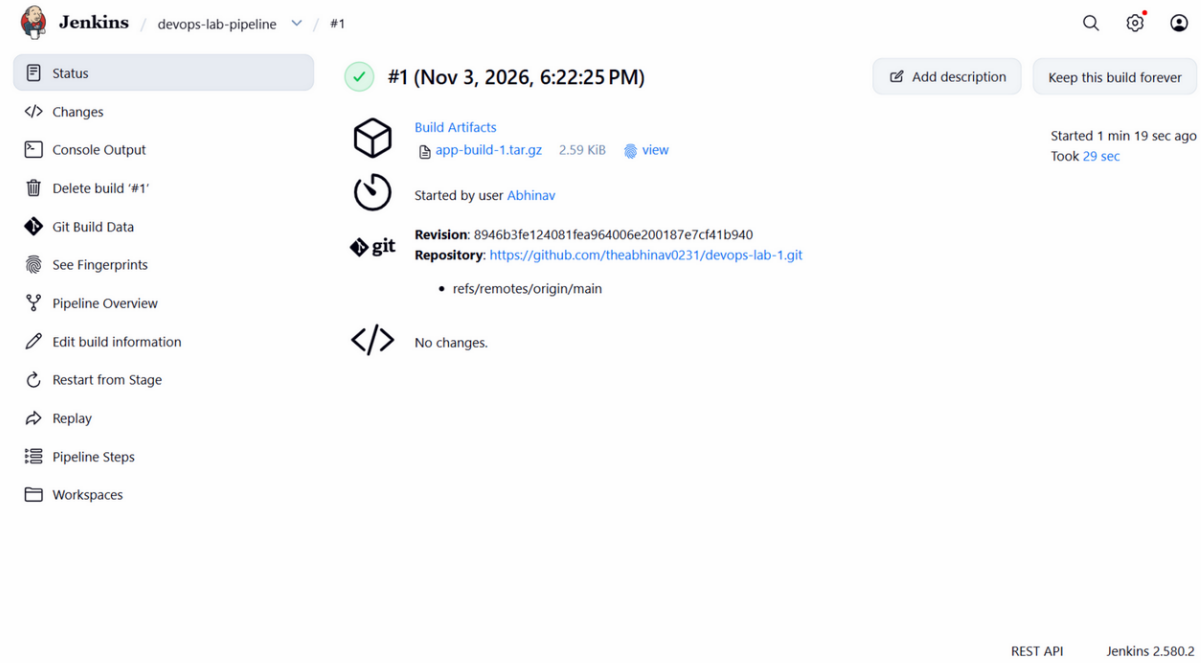


Fig: Stage graph of build #1 (all stages green)

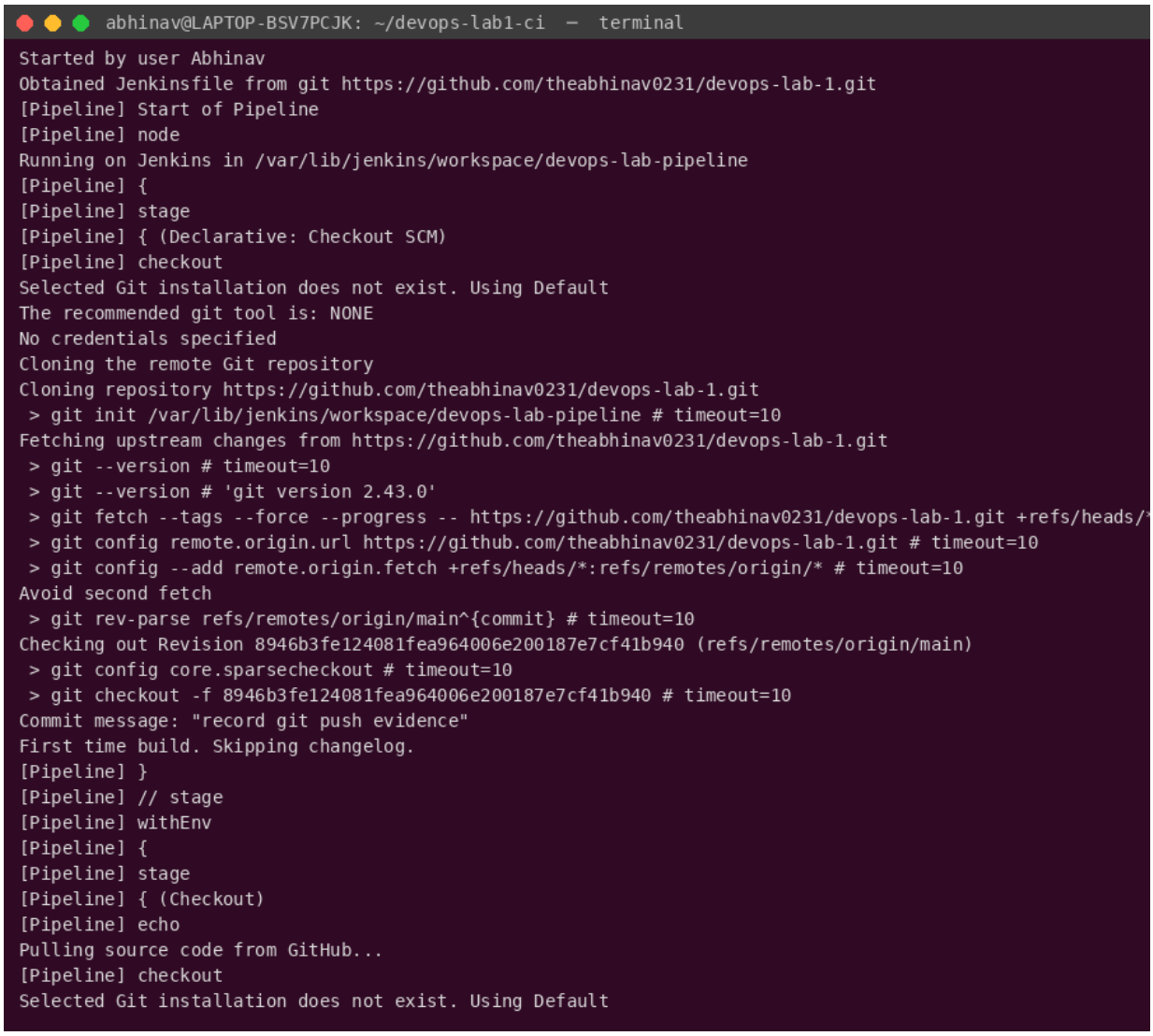


Fig: Build #1 console output (part 1: checkout and build)

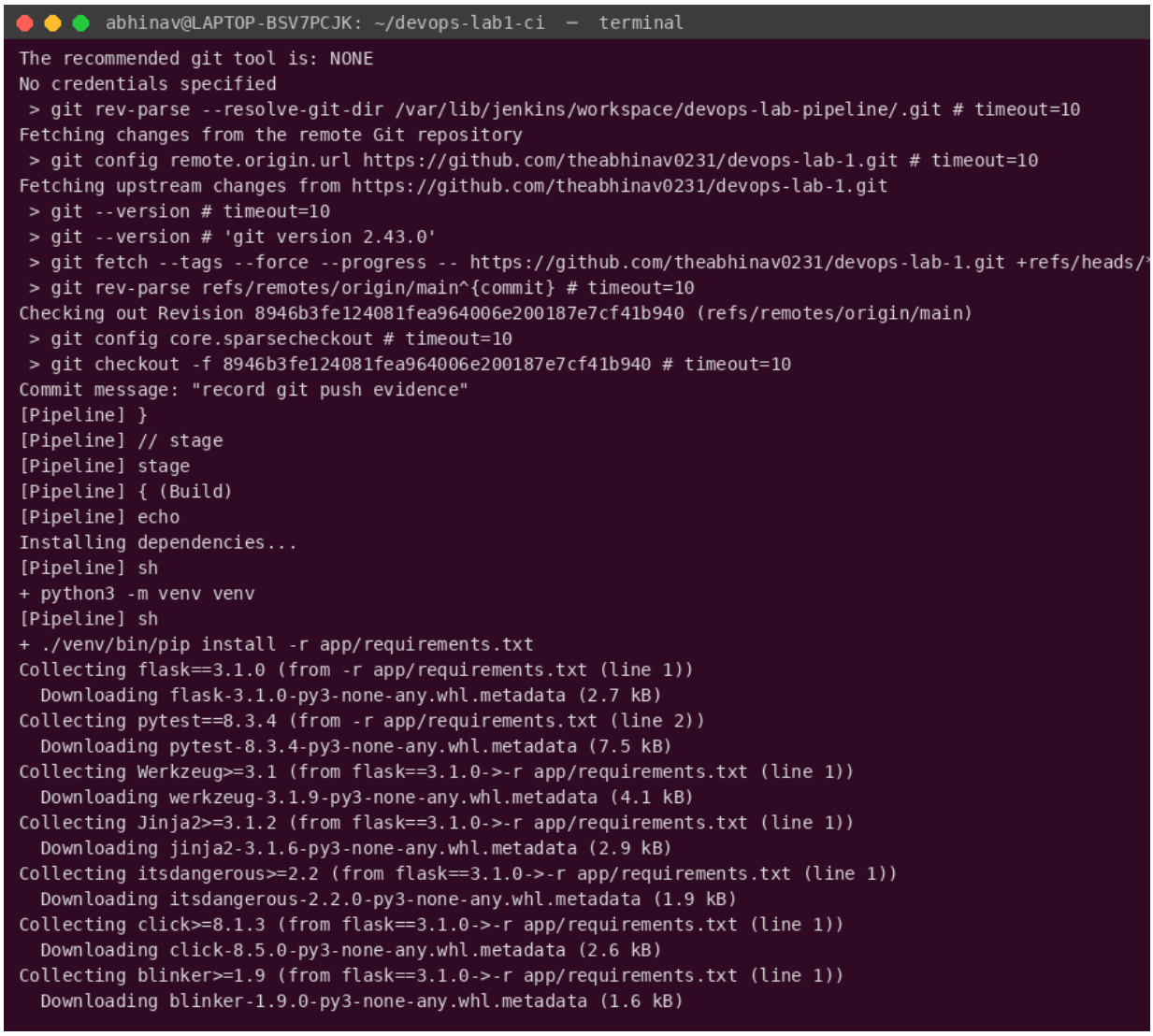


Fig: Build #1 console output (part 2: tests passing)

Observations

- Because the Jenkinsfile lives in the repository, the pipeline definition is version controlled together with the code.
- Lightweight checkout reads the Jenkinsfile before a full clone, so pipeline changes take effect immediately.
- The artifact app-build-1.tar.gz is archived and fingerprinted on every build.

Conclusion

All the commands were executed successfully on Ubuntu 24.04 LTS and the evidence (command logs, screenshots and commit history) is committed in the GitHub repository github.com/theabhinav0231/devops-lab-1 under the docs/ directory.